

Deep Learning- Based Automated for Real-Time Heart Attack (Myocardial Infraction) Screening



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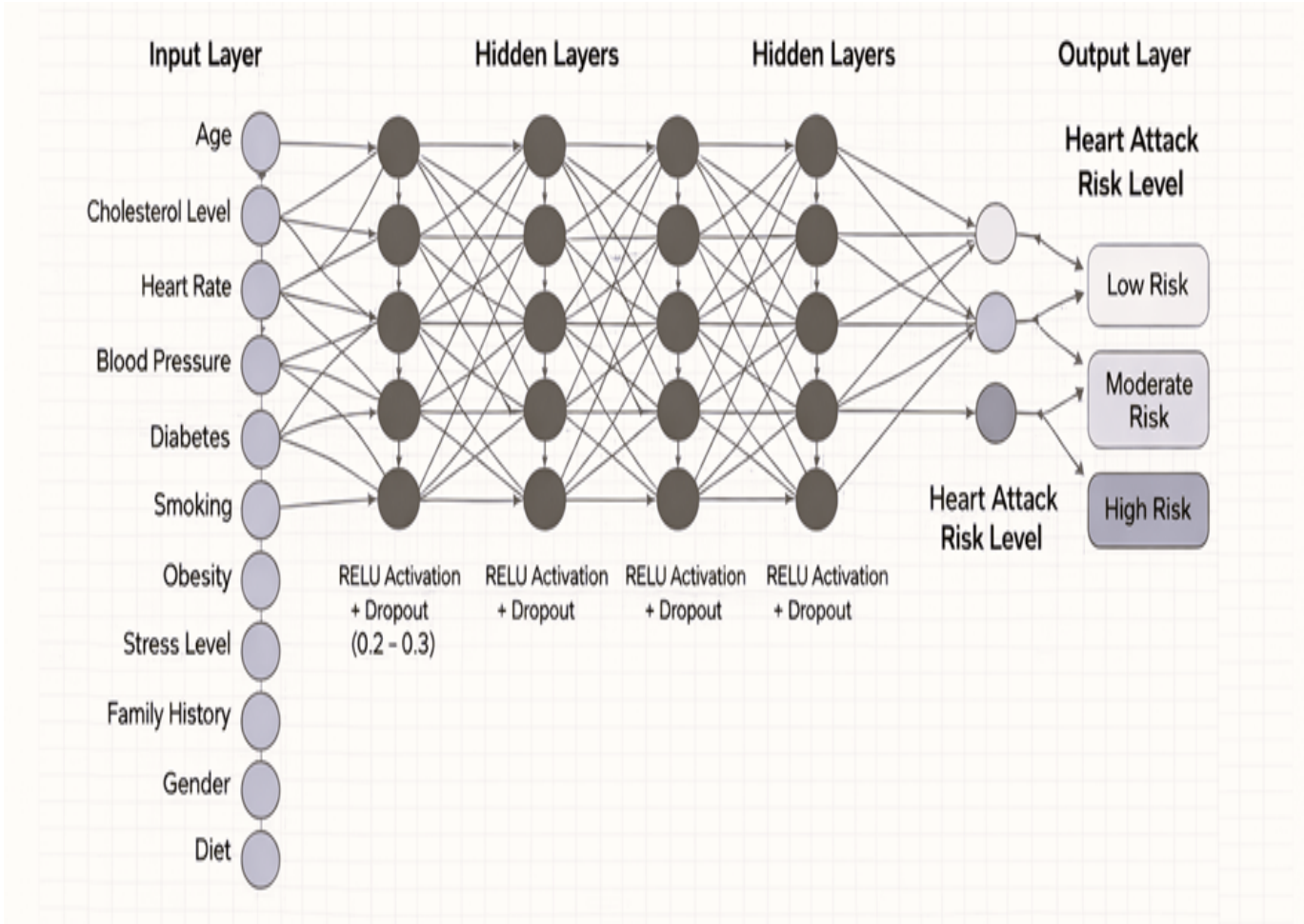
Problem & Motivation

Confusion Matrix

- **Problem Context:** Inaccurate heart disease risk assessment stems from complex health data, imbalanced high-risk datasets, and poor accessibility to early diagnostic tools.
- **Research Gap:** There is a lack of an automated, accessible, multi-level heart disease risk screening systems hat can accurately process complex multi-factor clinical/ lifestyle data while reliably detecting rare, high-risk patients in imbalanced datasets.
- **Objective:** Develop an automated heart attacking screening system using a **Multi-Layer Perceptron (MLP)** architecture to categorize patients into distinct risk levels based on clinical and lifestyle attributes

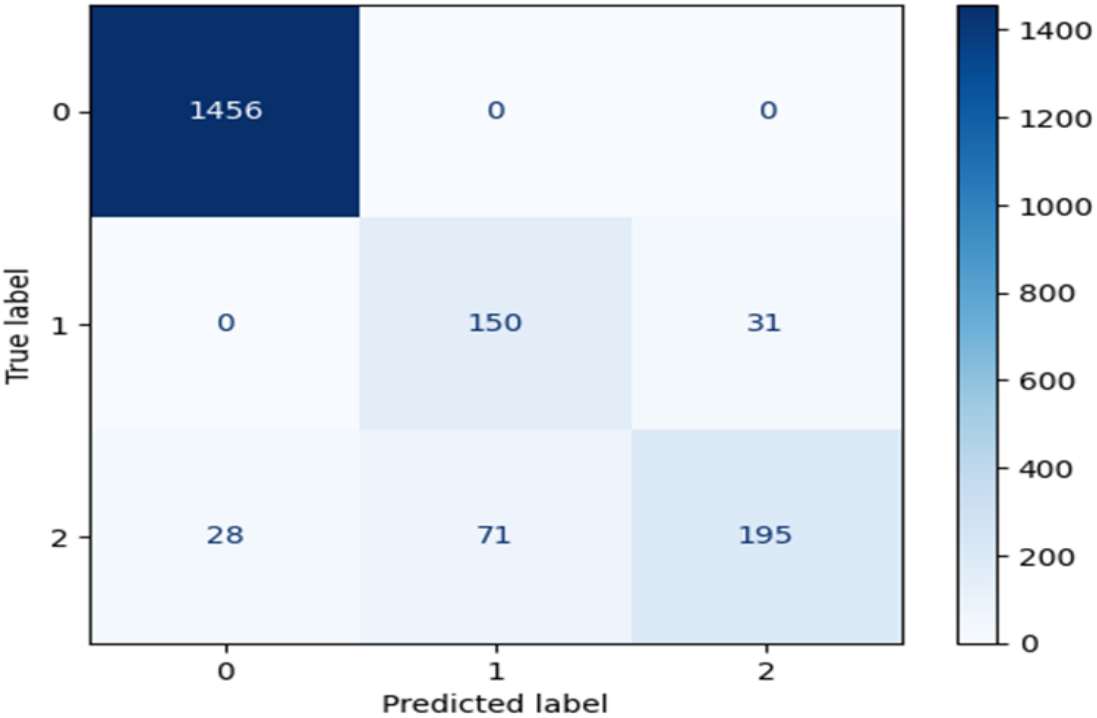
Methodology

System Architecture
Our system utilizes a **Multilayer Perceptron (MLP)** network for heart attack risk classification. This architecture is well-suited for processing tabular numeric data and mapping multi-variable patient metrics into discrete risk categories.



Evaluation & Key Results

Our MLP achieved **93% test accuracy**. Confusion matrix is show below:



Dataset

Our data is sourced from Kaggle Garbage Classification Dataset consisting of: (15,150+ images, 12 waste classes).

Age	Cholesterol	Heart rate	Diabetes	Heart Attack Risk
0.596	0.314	0.048	0	0
0.596	0.096	0.048	1	0
0.596	0.189	0.048	0	0
0.079	0.493	0.071	1	1
0.584	0.821	0.067	1	1
0.584	0.950	0.067	1	1

Significance & Conclusion

- **Contribution:** High-Accuracy Decision Support Tool: It delivers a trained model achieving a high test accuracy of 93.27% (and a weighted F1-Score of 0.93) to serve as a fast, data-driven diagnostic support system.
- **Impact:** Accessibility in Low-Resource Settings: Because the system utilizes a computationally lightweight MLP architecture processing structured tabular clinical data, it can be easily deployed in resource-constrained or rural.
- **Future Work:** Edge AI Optimization: Researching weight quantization and pruning techniques to allow the model to run efficiently on ultra-low-power, edge-computing devices ,Longitudinal & Advanced Data Integration , Addressing Class Imbalance:..